

Does Populism Influence Violence Against Journalists in Europe? Analysing Predictors, Typology, and Gendered Dimensions as Factors

Group Project V

June 2, 2026

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Executive Summary

This project explores the relationship between populist governance, media capture, and physical violence against journalists in Europe. Using the Mapping Media Freedom dataset's incident data amongst literature-informed variables, we create five Zero-Inflated Negative Binomial predictive models to estimate counts of physical violence against journalists. Control variables include poverty, military spending, and judicial independence, while other models use an altered version of Reporters Without Borders' media-capture index, seat-weighted populist presence from the PopuList and ParlGov datasets, as well as gender composition of journalist victims and the leading political cabinet. A feminist critical lens was applied to incorporate gender into the analysis, drawing on gender data available in existing datasets, while six semi-structured interviews provided additional qualitative insights.

Key Findings include:

- A one-standard-deviation increase in the populist share of a European government is associated with nearly fifty percent more physical violence counts against journalists.

- Adding populism variables improves the models' fit to seen data, but not unseen, making it unreliable for predicting.
- Government attacks on the judiciary only become significant after including populism, which links populist rule to general democratic failure
- Media capture, weighted populism, and gender composition add minimal predictive power, but instead add model explainability
- Weighted populism and its interaction with media capture sharpen the significance of institutional predictors and link populist governance and attacks on the judiciary to higher rates of violence against journalists.
- Interviews concur that political polarisation and outlet independence are generally more significant than the gender of a journalist.
- Partial populist presence in government is sufficient to correlate with higher rates of physical violence against journalists.
- Populism and government attacks on the judiciary pose a positive feedback loop with a compounding effect on journalist safety.
- Gender composition of the cabinet and the attacked journalists does not improve the prediction.

Weighted populism (model 2) is a positive and significant predictor (IRR = 1.51, 95% CI [1.15, 1.96], $p = .003$), with model 2 being the most predictive model. Crucially, populism does not displace the structural predictors but adds to them: relative poverty remains the dominant single predictor and barely moves from the baseline (IRR = 3.34, $p < .001$, versus 3.27 in Model 1). The introduction of populism also reshapes the institutional coefficients. Government attacks on the judiciary, non-significant in the baseline (IRR = 1.23, $p = .16$), becomes significant here (IRR = 1.39, $p = .024$), a one-standard-deviation rise associated with roughly 39% more incidents. The two security and institutional controls fade once populism enters: military expenditure drops from marginal significance to clearly non-significant (IRR = 1.11, $p = .43$), as does high court independence (IRR = 0.84, $p = .30$). Populist governance is limitedly linked to physical violence against journalists in Europe, with weak predictive power and no significant relationship to media capture nor the gender of journalists nor government.

Ultimately, incident reporting in countries with monitoring gaps can increase data-reporting and increased journalist protective initiatives should also pay heed to indirect physical harms, such as self-censorship and the chilling effect, to address conflict before it escalates into physical violence.

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Part I

Modelling Violence Against Journalists in Europe

1 Introduction

1.1 Press Freedom and the Rise of Populism in Europe

The relationship between populist governance and press freedom has emerged as one of the defining challenges of democratic politics. In Europe, governments with a communist background have been susceptible to democratic deficits and restricted human rights and have gained heavy influence by the introduction of populist parties, winning 36% of seats in the 2024 European elections (Gunnarsson and Zoega, 2017; ECPS, 2024). The scale of the threat to journalists increased from 82 incidents (killed, detained, hostage, or disappeared) in 1995 to 890 in 2025 (RSF, 2026). Additionally, the European Center for Press & Media Freedom declared that “the share of media freedom violations per year in which political actors were responsible has increased from 14% in 2020 to 20% in 2025” (2026). Whether this surge in violence becomes a systemic threat remains an open question that this study seeks to address.

1.2 Europe as a Comparative Context

Populism is broadly defined as an ideology that frames politics as a divide between the “people” and the “corrupt elite” (Mudde, 2004). Europe provides a particularly interesting context for studying populism due to its diverse range of political trajectories, from liberal democracies in Western Europe to post-communist states in Central and Eastern Europe where authoritarian legacies continue to shape political culture. Wettstein et al. (2018) highlights that the relationship between media and populists is conditioned by the “cordon sanitaire.” In Western democracies, such as Germany and Sweden, mainstream parties often exclude radical parties from governing coalitions, with journalists similarly marginalizing populists as “laughable” or “dangerous.” In Central and Eastern Europe (e.g., Hungary, Poland, Bulgaria), where populists often govern, independent media are instead portrayed as the “lying press” or “fake news.” Here, the media may reproduce populist narratives to survive financially. This variation makes Europe an ideal comparative setting for understanding the relationship between populism and journalist safety.

Physical violence against journalists in Europe is increasingly linked to the political environment in which journalists operate (Fawzi and Krämer 2021, cited in Wolleschensky & Sehl 2025). Such violence generates a chilling effect that strikes the ability of journalists to act as a watchdog and scrutinize power precisely when it most needs scrutinizing, in which the ability for freedom of speech becomes suppressed. This system is moderated by media capture, occurring when press autonomy is weakened by political interests, ultimately limiting accountability and strengthening populist power.

1.3 Research Question

This study therefore asks:

To what extent does populist governance predict physical violence against journalists in Europe, and is this relationship conditioned by the degree of media capture in the country?

This question is addressed by modelling violence against journalists, including physical attacks, examining the extent to which populist governance predicts such incidents in Europe.

2 Theoretical Background

2.1 Defining Populism

The definition of populism is open and contested by many scholars. Some argue populism should be mainly understood as an ideology, while others argue it is more a political style or discourse. This conceptual ambiguity has produced a broad and often inconsistent body of literature, making it important to establish a clear definition at the outset of this study.

This paper adopts the definition proposed by Mudde (2004), which has become one of the most influential conceptualisations within contemporary populism research:

“a thin-centred ideology that considers society to be ultimately separated into two homogeneous and antagonistic groups, ‘the pure people’ versus ‘the corrupt elite’.”
(Mudde, 2004, p. 543)

Mudde notes that populist ideology inherently relies on a Manichean outlook (Mudde, 2004). Named after the ancient religious philosophy of Manichaeism, which divides the world strictly into good and evil, populists view politics as a stark, uncompromising struggle between the “righteous people” and the “corrupt elite.”

A thin-centred ideology contains only a limited set of core ideas and therefore cannot, on its own, provide a complete political worldview or programme for governing society, unlike “thick” ideologies such as liberalism, socialism, or conservatism. Because populist principles alone do not explain how society or the economy should be organised, Mudde and Kaltwasser (2017) argue that populism must attach itself to other, more comprehensive ideologies. In contemporary politics, populism can therefore emerge across both right- and left-wing political agendas.

2.2 Populism and the Media

Populist antagonism towards the media can be understood as a direct extension of this Manichean divide. This is captured in what scholars call antimedia populism. This term refers to the belief that journalists are not neutral actors, but part of a self-serving elite that are detached from ordinary people and their interests (Fawzi and Krämer 2021, cited in Wolleschensky & Sehl 2025).

A key assumption is that the media does not merely report on politics, but actively distorts it in favour of those in power. As Krämer (2018, cited in Relly 2021) puts it, the populist perspective on the news media is that “liberal journalism betrays the people and conspires with, or is instrumentalised by, the ruling elite to manipulate the people.” In this view, the media is seen as “against the people,” rather than as an independent institution.

This way, delegitimising media criticism becomes a key expression of antimedia populism, going beyond normal scrutiny. Instead of offering reasoned feedback, it often relies on accusations, emotional language and claims that journalists are biased or dishonest (Egelhofer, Aaldering, and Lecheler 2021, cited in Wolleschensky & Sehl 2025). Because such criticism is aimed at undermining trust in journalism itself, scholars describe it as “populist media criticism” (Fawzi 2020).

2.3 The Effect of Populist Violence on Journalists and Democratic Accountability

Anti-media populism does not remain at the level of rhetoric. Delegitimation creates the permissive conditions for hostility to escalate, moving from verbal attacks and online intimidation

toward physical harm (see Appendix A: *Chain of Violence – How Populist Regimes Target Journalists*). Although verbal and online incidents make up the bulk of recorded violations, nearly one in five (19%) are physical attacks (MFRR, 2026). Physical violence is the most direct expression of this hostility as it targets journalists' bodily integrity and their capacity to perform a democratic watchdog function, and as such marks the most extreme end of the threats to press freedom.

The consequences of violence against journalists extend beyond individual victims. A key outcome is the so-called "chilling effect": a gradual withdrawal from reporting on sensitive or controversial topics due to fear of harassment, threats, or attacks. Kim and Shin (2022) document how journalists subjected to anti-press violence respond not by pushing back, but by turning inward. With little institutional support from their newsrooms, many adopt emotion-focused coping strategies to manage their psychological stress rather than confronting it. As a result, journalists who are managing fear and exhaustion might be more inclined to self-censor or are less likely to pursue the confrontational, accountability-driven reporting that populist leaders most want to suppress.

The result is a feedback loop that benefits those in power. A more cautious press gives populist governments greater room to operate without scrutiny, which in turn consolidates their power and emboldens further attacks on the journalists who remain. The chilling effect is therefore not a side effect of anti-press violence, rather it is its purpose.

2.4 Media Capture and Press Freedom

Media capture refers to a situation in which media organisations are unable to operate independently because they are influenced by political or economic actors with vested interests (Schiffrin, 2017). Unlike overt censorship, media capture operates through more indirect mechanisms, such as ownership structures, financial dependencies, favourable regulation, or the distribution of state resources. For example, in the early 2000s, the Social Democrat (PSD) government in Romania needed to appear democratically reformed for EU accession while maintaining domestic control. Leaked government transcripts revealed they discussed using state advertising and tax breaks as a way to buy a "vague, individual reprieve" from critical coverage (Pippidi, 2008).

As a result, media outlets become less able to fulfil their democratic role as independent watchdogs that hold political actors accountable (Schiffrin, 2017). Instead, they may prioritise the interests of political and economic elites over the public interest. Mungiu-Pippidi (2008) argues that media capture in Europe is rarely absolute and often functions as a "companion of state capture," reflecting broader networks of political influence.

Media capture is particularly relevant in the context of populism. When independent journalism is weakened, populist attacks on the media may face less resistance and scrutiny. By reducing accountability and limiting the plurality of voices within the media landscape, media capture can create conditions in which hostility towards journalists and broader threats to press freedom are more likely to emerge.

2.5 Contextual Factors

This study includes a set of structural and institutional variables that capture the broader environment in which violence against journalists occurs. Rather than functioning as purely technical controls, these factors are grounded in established literature on press freedom, state capacity, and democratic stability, and are included because they shape the underlying conditions

in which political actors operate. These include military expenditure, economic development and inequality, and judicial independence (Brambila, 2017; D'Agostino, Dunne, & Pieroni, 2012; Shin, 2020; Schwuchow, 2023; Aydin-Cakir & Akarçay, 2025), all of which are further explained in Appendix B: *Contextual Factors*. Economic development and inequality reflect socioeconomic conditions that may increase vulnerability to violence and organised crime (Brambila, 2017; Schwuchow, 2023). Judicial independence measures the capacity of legal institutions to protect journalists and hold perpetrators accountable, while attacks on the judiciary indicate broader processes of institutional erosion (Brambila, 2017; Aydin-Cakir & Akarçay, 2025).

3 Methodology

To answer our research question, three regression models were developed and compared (see Appendix C for *Data Flow Diagram* and Appendix D for *Model development details*).

3.1 Outcome Variable: Physical Attacks Against Journalists

The dependent variable is the number of physical attacks against journalists recorded by the Mapping Media Freedom (MMF) database. MMF documents violations of media freedom across Europe and classifies incidents according to their nature. For this study, only physical incidents were included. These comprise physical assaults with or without injury, deaths resulting from physical attacks, bombings, shelling, explosions, arson attacks, abductions or kidnappings of journalists, and sexual assaults. By focusing on physical violence, the analysis captures the most severe forms of intimidation directed at journalists while relying on a consistently reported outcome measure across countries and years.

3.2 Model 1: Baseline Model – Control Variables

The first model serves as a baseline against which the explanatory value of populism and media capture can be evaluated. Based on the theoretical framework and existing literature, a selection of contextual variables was made: government attacks on the judiciary, high court independence, military expenditure, and the relative-poverty headcount. These variables capture institutional, political, and socioeconomic conditions that may influence violence against journalists.

The final baseline model includes four control variables:

- *Government attacks on the judiciary*, were measured using the Varieties of Democracy (V-dem) indicator Government Attacks on the Judiciary (v2juhccattack). This variable captures the frequency with which government officials publicly question the integrity of the judiciary by portraying courts as corrupt, incompetent, or politically motivated. The indicator is based on expert assessments and ranges from frequent attacks to the absence of attacks. Higher values indicate fewer government attacks on judicial institutions and therefore stronger respect for judicial independence.
- *High court independence* uses the V-Dem (2026) indicator. This variable assesses the extent to which high courts make decisions independently of government preferences in politically salient cases. Higher values indicate greater judicial independence and lower levels of political influence over the judiciary.
- *Military expenditure* as a percentage of GDP, sourced from the World Bank (2022), reflecting the share of national economic output allocated to military spending.

- *Relative poverty*: Number of people below 60% of the median, obtained from Our World in Data (2025), measuring the proportion of individuals living below 60% of median income.

3.3 Model 2: Baseline Model Plus Weighted Populism

The second model introduces populism as the primary explanatory variable. This model directly addresses whether populist participation in government contributes to the prediction of physical attacks against journalists beyond the effects of the baseline controls. If populism contributes meaningfully to explaining variation in journalist violence, Model 2 should demonstrate improved predictive performance relative to the baseline model.

3.3.1 Calculating Weighted Populism

Populism is operationalised using data from two sources. The PopuList 3.0 dataset (Rooduijn et al., 2023) identifies political parties in Europe that can be classified as populist according to Mudde’s definition of populism (see *Appendix E: Annotated Data Sources*). This conceptualisation aligns with the theoretical framework of the present study. Populist 3.0 assigns each party represented in government a value of 1 if classified as populist and 0 otherwise. Second, the ParlGov database (Döring Manow, 2024) provides the number of seats each party holds in government and the total number of seats in cabinet in a given year across 33 European countries, ranging from 2015 to 2023.

A weighted measure was constructed to capture variations in the strength of populist representation:

$$\text{Weighted Populism}_{c,t} = \frac{\sum_{p \in \text{populist}} \text{seats}_{p,c,t}}{\text{total seats}_{c,t}}$$

Each populist party that has a cabinet position in a given country-year, ppopulist , is multiplied by the party’s share of all legislative seats, seat_p , which is then divided by the number total seats in the parliament to determine the weighted populism score of country, c , in a given year, t . This measure ranges from 0, indicating that no populist parties hold governing positions, to 1, indicating that all governing seats are occupied by populist parties. The weighted approach captures gradations in populist influence and reflects the coalition-based nature of many European governments.

3.4 Model 3: Baseline Model + Populism + Media Capture

The third model examines whether media capture moderates the relationship between populism and violence against journalists. Media capture is proxied using the World Press Freedom Index produced by Reporters Without Borders (RSF, 2026), which evaluates national media environments across political, economic, legal, and social dimensions. To avoid circularity with the dependent variable, the incident-based safety component of the index is excluded. Media capture is included solely as an interaction term with weighted populism, allowing us to test whether the effect of populist governance on violence against journalists depends on the level of media capture, with stronger effects expected in more captured media environments.

The remaining structural indicators were transformed into a media capture variable, by inverting the original press freedom score:

$$\text{Media Capture} = 100 - \frac{\text{RSF Score}}{100}$$

The resulting variable ranges from 0 to 1, where 0 represents complete press freedom and 1 indicates maximum media capture. Higher values therefore correspond to weaker media independence and greater opportunities for political influence over media institutions.

3.5 Model Choice

Given the structure of the dependent variable, a Zero-Inflated Negative Binomial (ZINB) model is employed. The outcome is a count variable characterised by overdispersion and a large proportion of zeros. A standard Poisson model would underestimate variance and produce biased standard errors, while a standard Negative Binomial model does not explicitly account for excess zeros. The ZINB specification therefore provides a more appropriate fit by modelling both the count process and the excess-zero generating process separately.

4 Results

4.1 Model 1: Baseline Model

The baseline model tests whether the structural control variables alone predict incidents of physical violence against journalists. Table 1 reports the results.

Table 1: Model 1: Baseline model — coefficients, IRRs and predictive performance ($N = 174$, 7 parameters).

Component & Coef. & IRR & <i>p</i> -value			
<i>Incidence Rate Ratios</i>			
Relative poverty (below 60% median)	1.18	3.27***	< 0.001
Military expenditure (% GDP)	0.21	1.23 [†]	0.096
Government attacks on judiciary	0.21	1.23	0.162
High court independence	-0.22	0.80	0.211
Constant	0.36	1.44**	0.002
<i>Model fit (full-data refit)</i>			
AIC		623.08	
BIC		645.19	
McFadden R^2		0.749	
<i>5-fold cross-validation (mean $\pm$ SD)</i>			
RMSE		8.12 $\pm$ 2.27	
Bias		-0.09 $\pm$ 1.27	
Predicted-observed corr.		0.66 $\pm$ 0.09	

Note. Zero-Inflated Negative Binomial estimator. *** $p < 0.001$; ** $p < 0.01$; [†] $p < 0.10$.

Even without any populism measures, the baseline captures the broad cross-national pattern of journalist violence; its prediction error (RMSE of 8.12 ± 2.27) is slightly lower than the outcome's own standard deviation of 10.98. However, its out-of-sample predictions are only slightly better

than a naive model that always predicts the average outcome (R^2 of 0.01 ± 0.87). Relative poverty is the strongest predictor on the full-data refit (coefficient & IRR) ($IRR = 3.27$, $p < .001$) (see table 1). Approximately 3.25 times the anticipated number of physical incidents is linked to a one-standard-deviation rise in the poverty population. This baseline model demonstrates that, prior to the introduction of any populist measures, a country's institutional and socioeconomic structure already explains some portion of the difference in journalist violence, though it does not predict well.

4.2 Model 2: Weighted Populism

Model 2 adds weighted populism as a direct predictor (Table 2).

Table 2: Model 2: Baseline + weighted populism — coefficients, IRRs and predictive performance ($N = 174$, 8 parameters).

Component	Coef.	IRR	<i>p</i> -value
<i>Incidence Rate Ratios</i>			
Relative poverty (below 60% median)	1.21	3.34***	< 0.001
Weighted populism	0.41	1.51**	0.003
Government attacks on judiciary	0.33	1.39*	0.025
High court independence	-0.17	0.84	0.302
Military expenditure (% GDP)	0.10	1.11	0.429
Constant	0.32	1.37**	0.0064
<i>Model fit (full-data refit)</i>			
AIC		615.31	
BIC		640.59	
McFadden R^2		0.753	
<i>5-fold cross-validation (mean $\pm$ SD)</i>			
RMSE		9.49 $\pm$ 1.69	
Bias		0.46 $\pm$ 2.02	
Predicted–observed corr.		0.67 $\pm$ 0.03	

Note. Zero-Inflated Negative Binomial estimator. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Adding populism adds little predictive power as model 2 returns an RMSE of 9.49 (± 1.69), with a slight inclination to overpredict (bias 0.46 ± 2.02) (see table 2). However, weighted populism is a significant predictor, with one-standard-deviation rise in the seat-weighted populist share in government is associated with approximately 50% more predicted physical incidents ($IRR = 1.51$, $p = .003$) (see figure 5). Government attacks on the judiciary become significant here ($IRR = 1.39$, $p = .024$), a one-standard-deviation rise associated with roughly 39% more incidents, a shift we return to in the model comparison.

4.3 Model 3: Weighted Populism $\times$ RSF Interaction

Model 3 uses weighted populism $\times$ RSF press-freedom interaction, testing whether the effect of populism is conditioned by the degree of media capture (Table 3).

Table 3: Model 3: Baseline + populism $\times$ RSF interaction — coefficients, IRRs and predictive performance ($N = 174$, 8 parameters).

Component	Coef.	IRR	<i>p</i> -value
<i>Incidence Rate Ratios</i>			
Relative poverty (below 60% median)	1.20	3.33***	< 0.001
Populism $\times$ RSF	0.37	1.45*	0.016
Government attacks on judiciary	0.31	1.36*	0.036
High court independence	-0.13	0.88	0.481
Military expenditure (% GDP)	0.07	1.07	0.637
Constant	0.34	1.40**	0.004
<i>Model fit (full-data refit)</i>			
AIC		618.70	
BIC		643.97	
McFadden R^2		0.752	
<i>5-fold cross-validation (mean $\pm$ SD)</i>			
RMSE		9.32 $\pm$ 1.49	
Bias		0.41 $\pm$ 1.77	
Predicted–observed corr.		0.67 $\pm$ 0.05	

Note. Zero-Inflated Negative Binomial estimator. *** $p < 0.001$; * $p < 0.05$.

Out-of-sample, Model 3 performs much like Model 2 (RMSE of 9.32 ($\pm$ 1.49)) (see table 3). On the full-data refit, the association between populist governance and physical violence is stronger where media capture is higher as the interactive term variable is a positive and significant predictor (IRR = 1.45, $p = .016$) (see table 3). Relative poverty remains the dominant predictor (IRR = 3.33, $p < .001$) and government attacks on the judiciary stays significant (IRR = 1.36, $p = .036$), while military expenditure (IRR = 1.07, $p = .64$) and high court independence (IRR = 0.88, $p = .48$) remain non-significant, as in Model 2.

See Appendix F: *Description of results* for more detailed result interpretations of the models here and in the Modelling Comparison section.

4.4 Model Comparison

Table 4 presents all three models side by side. Model fit is assessed using AIC and BIC (lower = better), McFadden R^2 , and out-of-sample RMSE from 5-fold cross-validation.

Table 4: Comparison of Models 1–3: IRRs and fit statistics. All models estimated on $N = 174$ country-years using a Zero-Inflated Negative Binomial estimator.

	Model 1 <i>Baseline</i>	Model 2 <i>+ Populism</i>	Model 3 <i>+ Pop. $\times$ RSF</i>
<i>Incidence Rate Ratios (IRR)</i>			
Relative poverty	3.27***	3.34***	3.33***
Weighted populism	—	1.51**	—
Populism $\times$ RSF	—	—	1.45*
Government attacks on judiciary	1.23	1.39*	1.36*
High court independence	0.80	0.84	0.88
Military expenditure (% GDP)	1.23 [†]	1.11	1.07
Constant	1.44**	1.37**	1.40**
<i>Model fit</i>			
AIC	623.08	615.31	618.70
BIC	645.19	640.59	643.97
McFadden R^2	0.749	0.753	0.752
<i>5-fold cross-validation</i>			
RMSE (mean $\pm$ SD)	8.12 $\pm$ 2.27	9.49 $\pm$ 1.69	9.32 $\pm$ 1.49
Bias (mean $\pm$ SD)	−0.09 $\pm$ 1.27	0.46 $\pm$ 2.02	0.41 $\pm$ 1.77
Predicted–observed corr.	0.66 $\pm$ 0.09	0.67 $\pm$ 0.03	0.67 $\pm$ 0.05
<i>Parameters & 7 & 8 & 8</i>			

Note. Bold values indicate the best-performing model on each metric. AIC and BIC: lower values indicate better fit. McFadden R^2 : higher values indicate better fit. RMSE: lower values indicate better out-of-sample predictive accuracy. Dashes indicate the term was not included in that specification.

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; [†] $p < 0.10$.

The cross-validation results in Table 4 indicate that adding weighted populism (Model 2) or its interaction with media capture (Model 3) does not improve out-of-sample predictive performance relative to the controls-only baseline (Model 1). Both models show higher RMSE (9.4875 ± 1.6914 and 9.3159 ± 1.4941 , respectively) compared to Model 1 (8.1199 ± 2.2659), indicating that the added predictors reduce out-of-sample predictive accuracy. Bias also increases with the introduction of the new predictors, with Models 2 and 3 consistently overestimating by 0.4618 ± 2.0244 and 0.4065 ± 1.7683 incidents. Model 2 achieves the lowest AIC (615.31) and BIC (640.59) compared to Model 1 (AIC = 623.08) and Model 3 (AIC = 618.70), suggesting that weighted populism improves model fit when penalising for complexity on the full dataset.

Consistently, adding weighted populism significantly improves the model’s ability to explain the observed data, as the Likelihood-ratio test yields significant results (M1 vs. M2: LR = 9.77, $p = .0018$), as does the full moderation model relative to controls alone (M1 vs. M3: LR = 6.38, $p = .0115$) (see figure 9). The addition of the interaction term alone (M2 vs. M3) does not yield a further significant improvement (LR = 3.38, $p = 1.000$), and the negative LR statistic suggests Model 3 fits the full data marginally worse than Model 2, despite the interaction term reaching significance in the coefficient table. This pattern, where LR tests favour more complex models, but cross-validation does not, is consistent with mild overfitting, meaning that the populism variables capture real variance in the observed data but do not generalise reliably to new observations.

Interestingly, government attacks on the judiciary become a significant predictor only upon the

introduction of the populism variables (see Table 4). This pattern suggests that populism and judicial interference may tap into a shared underlying dimension of democratic backsliding, and that controlling for populism clarifies the independent contribution of institutional attacks to journalist violence.

Although weighted populism and populism interaction variable (Populism x RSF) does not improve prediction, they are significant variables and also improve the model explainability of the data. They also improve the significance of government attacks on the judiciary, which links both populism variables with more corruption and non-democratic institutions. Together, these findings indicate that while weighted populism and its interaction with media capture do not improve out-of-sample prediction, they are substantively and statistically meaningful explanatory variables. They improve in-sample model fit, carry meaningful effect sizes, and appear to sharpen the significance of institutional predictors, linking populist governance and attacks on democratic institutions to higher rates of violence against journalists.

5 Discussion

While populism is a meaningful and significant explanatory variable, violence against journalists in Europe cannot be predicted solely based on how populist the governing cabinet is. It needs to be situated within the complex system of societal, economic, and institutional conditions that shape the environment in which journalists and politicians operate and interact (see Appendix G: *Stakeholder Analysis*). Consistently, relative poverty is reproduced as the dominant single predictor across every specification. This is consistent with Brambila (2017)'s finding that journalist violence concentrates in contexts of institutional weakness and economic deprivation and Schwuchow (2023), who links economic deprivation to the organised crime and corruption that heighten journalists' exposure to harm.

The positive association between populism and attacks on journalists matches the theoretical expectation that delegitimising hostility toward the press corresponds to greater violence against journalists, grounded in the antimedia-populism literature (Krämer 2018; Fawzi 2020). The sharpening of the judiciary-attacks coefficient once populism is added is consistent with Aydin-Cakir and Akarçay's (2025) account of governments targeting courts under conditions of corruption and contestation, in which populism and institutional interference move together as a cluster rather than independently.

Media capture failing to emerge as a strong moderator is itself consistent with Mungiu-Pippidi's argument that capture in Europe is rarely absolute and operates more subtly than the RSF proxy can detect. The media-capture interaction term, though significant on the full-data refit, added no out-of-sample predictive power and did not improve fit over Model 2, whereas the theoretical case for capture as a moderator would expect a stronger and more generalisable signal.

More broadly, the failure of out-of-sample prediction across all specifications suggests that populism and media capture alone cannot forecast violence against journalists, which slightly contradicts the theoretical case for populism and media capture as drivers above. Additionally, the two evaluation methods also point in opposite directions as likelihood-ratio tests favour the populism models, while cross-validation does not. This pattern is consistent with mild overfitting, where the populism variables capture real variance in the observed data but do not generalise reliably to new cases. As such, populism demonstrates explanatory power within the observed data but lacks the generalisability required for reliable prediction.

The fact that populism emerges as a significant explanatory variable even in a European context, where fully populist cabinets are rare, is itself a concerning finding. It suggests that even partial populist presence in government is sufficient to correlate with measurably higher rates of physical violence against journalists. Given that populist parties are gaining more seat share across multiple European countries (Ivaldi & Zankina, 2024), the conditions our model identifies as explanatory are becoming more rather than less prevalent. Additionally, the mutual reinforcement between populism and government attacks on the judiciary points toward a potentially compounding effect on journalist safety. Taken together, these points show that physical violence against journalists in Europe cannot be reliably predicted from how populist the governing cabinet is. Populism is statistically significant yet predictively weak, meaning that it is relevant for explaining the data, unreliable for forecasting it. However, when situating it in the wider context of the mutually reinforcing way populism, institutional erosion, and structural deprivation operate together, it remains a meaningful source of concern for the safety of journalists, particularly as populist parties continue to gain ground across Europe.

Part II

Bias Audit of the Model and Research Design

6 Questioning Model Assumptions

6.1 Omitted Variable Bias and Gendered Political Culture

The predictive model in this study assumes that populism and media capture are key predictors of threats and violence against journalists. However, this specification rests on the implicit assumption that governments can be treated as gender-neutral institutions, rather than as politically and culturally situated actors shaped by who holds power.

Excluding cabinet gender composition from the model risks omitted variable bias. If gender composition is correlated with both populist governance and levels of hostility towards journalists, then the estimated effect of populism may partially capture underlying gendered political dynamics rather than populist rhetoric itself.

Tellingly, hostile environments for press freedom may not be driven exclusively by populist ideology, but also by masculine political cultures embedded within government leadership structures. Gender composition should therefore be understood not merely as a demographic variable, but as a proxy for broader cultural norms shaping rhetorical styles, tolerance of dissent, and attitudes towards journalistic scrutiny.

6.2 Representation Bias

Representation bias arose from the likelihood that chilling effects and self-censorship render certain experiences, particularly those of journalists in hostile environments or women who have left the profession entirely, systematically invisible in the data and unable to reach. Additionally, our interview sample was also limited in its representational scope, capturing only a narrow slice of the diverse journalistic experiences that exist across Europe.

6.3 Annotation Bias

Annotation bias was mitigated through collaborative coding and shared definitions across researchers. However, interpretive variation between coders remained a limitation, particularly in classifying incidents across overlapping categories of hostility, intimidation, and violence.

6.4 Methodological Bias

The project's emphasis on gender introduces a *methodological trade-off*: while it allows for deeper engagement with gendered political dynamics, it also limits systematic attention to other potentially relevant axes of inequality, including race, migration status, and post-colonial power relations.

6.5 Framing Bias

Framing bias emerged in the operationalisation of core concepts: gender, violence, and populism each carrying definitional ambiguity that required ongoing reassessment throughout the analytical process.

6.6 Model Bias

Finally, *model bias* was identified in the risk of omitted variable bias and causal oversimplification, particularly the original assumption that governments could be treated as gender-neutral institutions, when masculine political culture may itself be a driver of hostile press environments independent of populist ideology.

Crucially, missingness in the data was acknowledged in discussion, treated not as a neutral technical problem but as a potentially meaningful indicator of structural inequality, demonstrating patterns of underreporting may themselves reflect the chilling effect the project set out to measure.

7 Gender as a confounding variable

Gender plays a central role in contemporary populist discourse and may shape how hostility towards journalists is expressed and legitimised. As Mostov (2021) argues, gender is embedded in many core populist narratives, particularly those centred around the protection of national identity, the family unit, religion, language, and culture, creating the gendered tropes and assumptions crucial to broader narratives of national recovery and social restoration.

Ideas of masculinity are often deeply embedded in populist leadership styles and rhetoric (Mostov, cited in Diallo 2024). Populist leaders frequently construct themselves as strong, “traditional” figures who promise to restore authority, stability, and social hierarchy, drawing upon national gendered tropes. This characterizes “patriarchal populism,” in which supporters position themselves in opposition to so-called “genderism” and progressive social change (Diallo 2024). Within these narratives, gender becomes part of a broader cultural struggle between “tradition” and “change” and “ordinary people” and perceived liberal elites, aligning with the simple populist dichotomy of “us versus them.” This often aligns with gendered oppositions such as “men versus women,” “families versus non-families,” or “tradition versus change” (cited in Diallo 2024).

Mostov further emphasises that populist discourse is not only gendered in its leadership styles, but also in its underlying logic of inclusion and exclusion. Populist actors often construct male “others” as threats, while simultaneously portraying “their” women as either vulnerable figures in need of protection or as symbols of national purity. This framing links gender directly to broader populist concerns with morality, belonging, and national identity, explaining why gender is not a peripheral issue but structurally embedded in populist narratives. Understanding this connection is also essential for recognising how populism can threaten women’s agency and undermine democratic institutions more broadly. If populist governments are also more likely to be associated with masculine political cultures and male-dominated leadership structures, then the relationship between populism and violence against journalists may partly reflect gendered political attitudes rather than populist rhetoric alone. In other words, journalists, particularly female journalists, may be targeted not only because they are portrayed as part of the “elite,” but also because critical journalism can be perceived as challenging traditional hierarchies of authority, masculinity, and social order, creating an environment that is implicitly hostile for journalistic safety.

7.1 Critical Research Question

Accordingly, this study also considers the following critical research question:

To what extent does the gender composition of governments shape distinct patterns of threat and violence against journalists in Europe?

8 Methodology: Semi-structured interviews

8.1 Semi-Structured Interview Design and Sampling Strategy

To complement the theoretical and model-based components of the project, we conducted six semi-structured interviews with journalists from different European contexts. The aim was to include a broad variety of journalistic backgrounds, work environments, and experiences with hostility or intimidation. Particular attention was given to the perspectives of female journalists, as the auditing specifically explores whether masculine political cultures shape threats and violence against journalists in gendered ways. Female journalists may experience these dynamics more directly, while male journalists may be less exposed to or less aware of certain forms of gendered hostility. For an overview of all participants, refer to Appendix H: *Participant Overview*.

Semi-structured interviews were chosen because they allowed participants to reflect openly on personal and potentially sensitive experiences while still providing enough structure to compare responses across interviews. This format was particularly suitable given the vulnerability of discussing violence, intimidation, harassment, and online abuse.

8.2 Interview Execution and Structure

The interviews asked participants to reflect on their own experiences and perceptions regarding the project's main themes: populism, media capture, and hostility towards journalists. Particular focus was placed on the role of gender and the gender composition of government. At the same time, the interview structure intentionally remained open-ended, allowing participants to reflect on other ways gender might influence experiences of intimidation, legitimacy, authority, or violence within journalism and political communication. For details of interview questions, see Appendix I: *Interview guide*.

Together, the interviews moved from descriptive accounts of personal experiences towards broader interpretive reflections on populism, gender, political culture, and press freedom. The qualitative component of the project was therefore used not only to gather empirical evidence, but also to critically evaluate the assumptions embedded within the broader research model.

9 Interview Results

Our interview results, through qualitative research, helped us answer the critical research question of whether the gender composition of governments shapes distinct patterns of threat and violence against journalists in Europe. Rather than confirming this directly, our interviews instead revealed why the model was unable to capture it, or at least provide more understanding. We structured our analysis across three levels: micro, meso, and macro, which can be referred to in Figure 1: Code Notewrok of Gendered Threats Against Journalists. Sifting one element of this system reshapes the others and the gendered dynamics at each level reinforce and reproduce one another. A full outline of all predictors across these levels is provided in Appendix J: *Bias Overview*. This overview is derived from our complete code analysis (see Figure 2, Frequency of Themes).

9.1 Micro-Level: Individual Experiences of Gendered Hostility

At the micro level, the most striking finding was that nearly all interviewees reported never having received explicitly gendered attacks. This sits in partial tension with the literature, with UNESCO

and OSCE documentation consistently identifying women journalists as disproportionately targeted by gendered and sexualised forms of hostility (Posetti & Shabbir, 2022), and the MMF data confirms that verbal and online attacks make up the majority of recorded incidents. The one exception was an interviewee who reported receiving sexual and hateful comments online.

9.2 Meso-Level: Organisational and Structural Conditions

At the meso level, our interviews did not surface clear evidence that funding structures affect women journalists differently than men in practice. This may partly reflect a limitation of our data collection. While our sample did include freelance and part-time women journalists, we did not explicitly ask about organisational structures, meaning no conclusions can be drawn. It may also reflect normalisation, where gendered organisational dynamics are so embedded they are no longer consciously registered. This sits in tension with the broader literature, which documents consistent gendered patterns in how work is organised. Women journalists are more likely to be concentrated in precarious, freelance, or short-term contracts, and are underrepresented in senior positions where decisions about coverage and resources are made (Robertson et al., 2021).

9.3 Macro-Level: Political Culture and Gendered Norms

At the macro level, our interviews revealed the relevance of masculine political culture and gendered political norms as a structural backdrop to how journalism is practised and perceived. One interviewee offered an example from Mexico, though not within Europe, is illustrative of the broader journalistic climate. She described not being taken seriously as a journalist in a context shaped by machismo, illustrating further how deeply embedded gender hierarchies can undermine the professional legitimacy of women journalists independently of any formal policy or legal framework. The same interviewee noted that this shifted meaningfully when a woman entered office with a pro-media stance in the UK, pointing to the symbolic significance of women in political leadership itself as a force that can reshape the cultural environment in which women journalists operate. This contradicts with our quantitative finding that gender composition of government was not a strong predictor for physical violence. Macro-level gendered dynamics operate through culture and norm-setting rather than through cabinet seat counts, which is precisely why they escape predictive power. This aligns with Mostov (2021) and Diallo's (2024) framework, which locates gendered power not in discrete institutional variables but in the broader cultural environment that political leadership both reflects and reinforces.

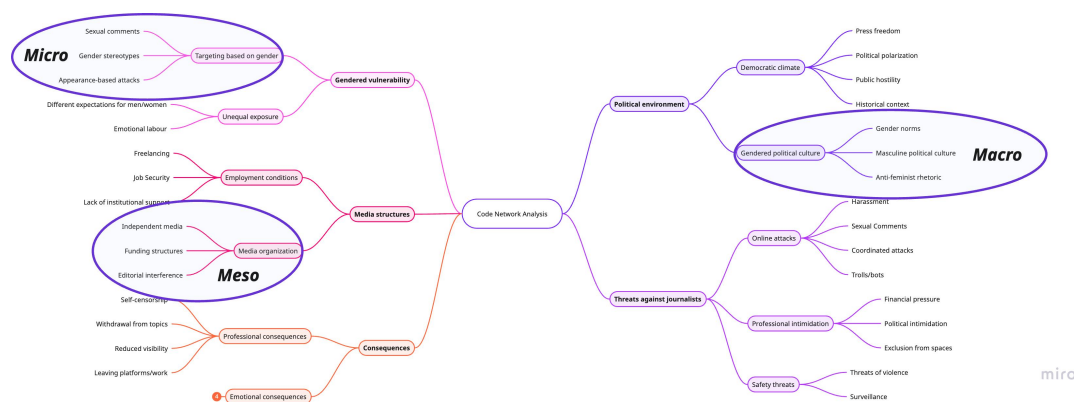


Figure 1: Code Network Analysis of Gendered Threats Against Journalists Across Micro-, Meso-, and Macro-Level Factors

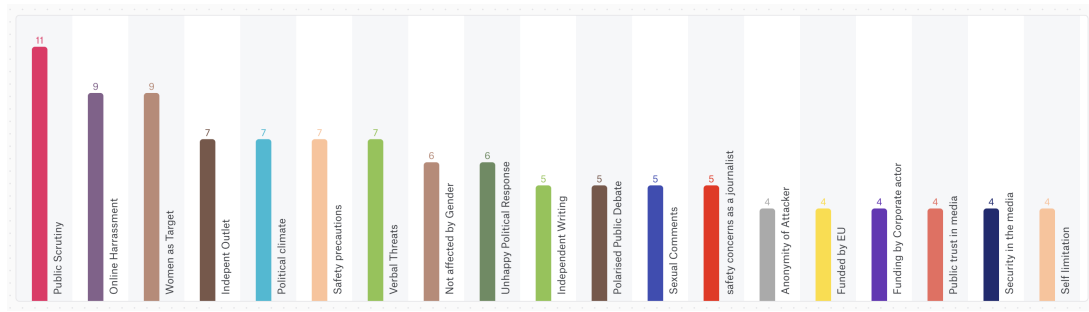


Figure 2: Frequency of Themes Identified in Interviews on Gendered Threats Against Journalists

10 Modeling Populism and Gender

To answer our critical research question we explored the populism model with added features: gender composition of government (model 4) and gender of journalist victims in the VDEM and MMF datasets (model 5). We compare this with model 2 as it has the lowest BIC (640.59) and AIC (615.31).

10.1 Model 4: Populism X Gender Composition in Government

This model adds V-Dem's "power distributed by gender" which represents how equal the distribution of political power is amongst gender, where the higher value, the higher levels of equality.

The variable, however, does not enhance the model and is non-significant ($p = .27$).

Table 5: Model 4: Populism x Gender composition government coefficients, IRRs and predictive performance ($N = 174$, 9 parameters).

Component	Coef.	IRR	p -value
<i>Incidence Rate Ratios</i>			
Military expenditure (% GDP)	0.08	1.08	0.539
High court independence	-0.25	0.78	0.173
Relative poverty (below 60% median)	1.19	3.29	< 0.001***
Government attacks on judiciary	0.28	1.33	0.0623 [†]
Weighted populism	0.41	1.50	0.0027**
Gender composition (power distribution)	0.18	1.19	0.270
Constant	0.31	1.36	0.008**
<i>Model fit (full-data refit)</i>			
AIC		616.09	
BIC		644.53	
McFadden R^2		0.754	
<i>5-fold cross-validation (mean $\pm$ SD)</i>			
RMSE		9.78 $\pm$ 1.87	
R^2		-2.28 $\pm$ 5.34	
Predicted-observed corr.		0.66 $\pm$ 0.02	
Bias		0.56 $\pm$ 2.05	

Note. Zero-Inflated Negative Binomial estimator. *** $p < 0.001$; ** $p < 0.01$; [†] $p < 0.10$.

This model is the weakest of the four main models. It yields the most over-prediction bias (0.56 ± 2.05), the lowest correlation, (0.66 ± 0.02), and the highest out-of-sample error (RMSE of $9.78 (\pm 1.87)$ and MAE of $4.09 (\pm 0.55)$ (Table 5). Additionally, the likelihood-ratio test against Model 2 is non-significant (LR = 1.22, df = 1, $p = .27$); weighted populism remains largely stable (IRR = 1.50, $p = .003$) and government attacks on the judiciary fall to marginal significance (IRR = 1.33, $p = .062$) (Table 5).

Taken together, it implies that once populist and structural constraints are taken into consideration, the overall gender balance of a government does not improve the predicted physical violence against journalists.

10.2 Model 5: Populism X Disaggregated Gender

Model 5 fits the Model 2 specification to physical events against female journalists (5b) and male journalists (5a).

Table 6: Model 5: Gender-disaggregated results — male vs female victims (coefficients, IRRs and p -values).

Component	Coef.	IRR	p -value
<i>Male victims</i>			
Military expenditure (% GDP)	0.0953	1.1000	0.5368
High court independence	-0.2474	0.7809	0.2643
Relative poverty (below 60%)	1.1285	3.0910	< 0.001
Government attacks on judiciary	0.5673	1.7636	0.0086
Weighted populism	0.4843	1.6230	0.0055
Constant	0.1547	1.1673	0.2929
<i>Female victims</i>			
Military expenditure (% GDP)	-0.1900	0.8270	0.3302
High court independence	-0.5347	0.5859	0.0412
Relative poverty (below 60%)	1.2924	3.6416	< 0.001
Government attacks on judiciary	0.5056	1.6580	0.0215
Weighted populism	0.2848	1.3296	0.0355
Constant	-1.4095	0.2443	< 0.001

Note. IRR = incidence rate ratio ($\exp(\text{coef})$).

Model 5 is not directly comparable to the others as it was fitted on one seventy/thirty hold-out instead of cross-validation. It revealed the effect is greater for male victims (IRR = 1.62, [1.15, 2.29], $p = .006$) than for female victims (IRR = 1.33, [1.02, 1.73], $p = .036$), while weighted populism is still a significant predictor in both cases. Relative poverty continues to be the primary predictor throughout, and government attacks on the judiciary are significant in both models (male IRR = 1.76, $p = .009$; female IRR = 1.66, $p = .022$). High judicial independence only becomes a significant protective factor in the female model (IRR = 0.59, $p = .041$). The male model predicts moderately well, with an RMSE of 7.83, an MAE of 3.20, a correlation of 0.65, and a slight under-prediction bias of 0.62. Contrastingly, the female sub-model returns smaller errors, an RMSE of 1.78 and an MAE of 0.80, with a correlation of 0.59 and an almost-zero bias of 0.04.

The female model's RMSE is significantly lower than the male model's (1.78 versus 7.83), but this is more likely due to the much smaller number of female occurrences than stronger prediction. The predicted–observed correlations are similar (male 0.66, female 0.60). Therefore, populism predicts violence against journalists of both genders, albeit somewhat more strongly for men in this sample, and that the quantitative data does not show a clear gendered mechanism when victim gender is broken down. This is consistent with the interview results, which showed that the journalist's gender was not as significant as we first hypothesized.

11 Discussion: Micro–Meso–Macro Dynamics

The bias hunt discussion ultimately revealed what the model alone could not capture, which is the lived, structural conditions that shape how safe journalists feel and what kinds of attacks they are exposed to (see Appendix J: Bias Overview for identified conditions that shape journalist's safety). Across our interviews, three factors emerged as most consequential, operating at a micro, meso, and macro level respectively.

11.1 Macro-Level: Gender Composition and Political Culture

At the macro-level, the gender composition of government, while theoretically relevant, did not prove significantly predictive (see table 5), suggesting that gendered political culture operates through more diffuse channels than cabinet composition alone can measure. The one instance where gender composition appeared potentially relevant was an British journalists interviewee's account from a job in Mexico, where the election of the country's first female president prompted speculation among locals about broader cultural shifts in how women are perceived, though this remained largely anecdotal, speculative, and outside the geographic scope of this study.

11.2 Meso-Level: Media Ownership and Institutional Constraints

At the meso-level, the ownership and independence of a journalist's outlet consistently shaped how freely they could work, confirming that institutional structure is a more immediate determinant of press freedom than national political context in isolation, as well as one that is itself shaped by the broader societal and cultural conditions specific to each country.

11.3 Micro-Level: Gendered Exposure and Individual Experience

11.3.1 Gendered Exposure to Hostility

At the micro-level, the gender of the individual journalist introduced additional layers of vulnerability, though notably, nearly all interviewees reported never having received explicitly gendered attacks. Where applicable, however, gender shaped the nature rather than the frequency of attacks, with gendered forms of hostility carrying a distinct psychological weight for the journalist, such as journalists taking extra precautionary measures before taking a job.

11.3.2 Interpretive Challenges in Incident Data

This finding also surfaces a deeper interpretive challenge within the MMF data itself. Male journalists account for 617 reported physical incidents compared to 148 for female journalists. Several structural factors may contribute to this fact, with women remaining underrepresented in political journalism. *Riedl, Rohrbach & Krakovsky (2022)* found that a journalist's own gender is the strongest predictor of their representation in political news coverage. The nature of attacks on women journalists may also skew more heavily toward verbal and online forms of violence rather than physical ones, and crucially, the chilling effect operates unevenly by gender (Posetti & Shabbir, 2022). If women journalists are more likely to self-censor, exit the profession, or avoid dangerous beats in response to threats, their lower physical incident counts may reflect not safety but structural missingness.

11.4 Cross-Level Factor: Geographic and Institutional Positioning

A further dimension that cut across all three levels was geographic and institutional positioning. A journalist based in Brussels operating within pan-European institutions inhabits a fundamentally different threat environment than one working in a more politically hostile or resource-scarce context, enabling journalistic freedom. This freedom was echoed by another journalist from the Czech Republic, who said herself that she would not be writing for her magazine if she had felt that her independence was compromised. Freedom, in this sense, is more an exception than a rule, contingent on a particular alignment of outlet independence, political environment, and personal circumstance.

11.5 Data Absence and the Chilling Effect

Perhaps most significantly, the bias hunt foregrounded the problem of what is missing from the data entirely. Self-censorship and the chilling effect mean that the most consequential outcome against journalists may never appear in any dataset, quietly hinting at the most violent state for journalists where it is not safe for them to speak.

12 Limitations

12.1 Predictive Performance and Explanatory Scope

The clearest limitation of this study is the modest predictive performance of all five models. Across specifications, cross-validated RMSE remained close to or above the outcome variable's standard deviation (10.98), with the baseline model reaching 8.12 (± 2.27) and populism models performing no better. In practice, this means the models predict physical incidents only marginally better than a naive average estimate. Although populism variables improve the observed-data fit, this explanatory power does not translate into out-of-sample forecasting, making the models better suited for explanation than prediction.

12.2 Data Constraints and Sample Limitations

A second set of limitations concerns sample size and the rarity of key cases. The panel is small, and two scarcities are especially consequential. First, physical incidents are relatively rare events, leaving few target cases for the model to learn from. Notably, none of our interviewees reported having been physically attacked, which may itself reflect a chilling effect in contexts where the risk of physical violence discourages journalists from coming forward or pursuing more exposed forms of reporting. Second, and most directly relevant to the research question, fully or majority-populist cabinets are uncommon in European politics during the study period, so the model has limited variance in its central independent variable to work with. A further simplification is the treatment of gender as a binary male/female variable in the gender models, which excludes non-binary journalists and politicians and cannot capture the full range of gendered dynamics the qualitative component points to.

12.3 Measurement, Temporal Structure, and Missingness

Data coverage and temporal assumptions impose a third set of constraints. The PopuList dataset ends in 2022, limiting the currency of the populism classifications relative to the rest of the panel. The one-year lag applied between predictors and incidents has only minimal theoretical justification; the relationship between political conditions and violence may operate on a different or variable timeline, which the fixed lag cannot accommodate. Relatedly, year-on-year changes in incident totals may coincide with discrete political events such as the election of a populist government or an episode of democratic backsliding, which a count model with lagged annual predictors is not well suited to capture. Missingness compounds these issues; as discussed in the methodology and bias-hunt sections, gaps in the data are not necessarily random and may themselves reflect the chilling effect and weaker monitoring infrastructure that the study set out to examine.

12.4 Qualitative Scope and Proxies

Finally, the qualitative component carries its own limitations of scope. Six interviews capture only a narrow slice of the diverse journalistic experiences across Europe, and the interview selection

was shaped by access and availability rather than systematic sampling. As noted in the bias audit, this leaves certain experiences underrepresented, particularly those of journalists who have left the profession or who self-censor. The limited predictive power may also reflect inadequate proxies. For example, the weighted cabinet populism score captures structural seat share but may miss rhetorical intensity or governing style, which could more directly drive hostility toward journalists. It may also indicate that structural factors such as poverty, court independence, and military expenditure are more stable predictors of violence, suggesting institutional conditions matter more than cabinet ideology in a given year. Alternatively, populism theory may simply lack predictive utility, as it was developed to explain political phenomena rather than forecast incident counts. A further limitation is sample size: full or majority-populist cabinets were rare in Europe during the study period, limiting variation in the key independent variable. Finally, the use of a fixed lag between political conditions and violent incidents may oversimplify a relationship that operates on a variable timeline, further weakening out-of-sample predictive power.

13 Conclusion

This study asked whether populist governance predicts physical violence against journalists in Europe. The answer is only as part of a wider, mutually reinforcing system, alongside structural predictors like poverty and judicial erosion which are compounded in predictive power strength when populism is included. When populist governance and attacks on the judiciary rise together, the threat environment for journalists does not worsen additively but risks doing so exponentially, each force amplifying the other. Gender, meanwhile, operates through culture and institutional norms rather than cabinet composition.

Populism emerges as a meaningful explanatory variable even in a European context where fully populist cabinets remain rare, suggesting that partial populist presence is sufficient to correlate with measurably higher rates of violence against journalists. Given that populist parties are gaining ground across Europe, the conditions this study identifies are becoming more prevalent.

Future research should seek to disentangle the temporal sequencing of these mechanisms, specifically whether judicial erosion precedes and enables media capture or whether both are driven simultaneously by populist entrenchment, and incorporate qualitative case variation that incident-level data alone cannot capture. Methodologically, there is also a need to improve the consistency and completeness of violence reporting across European countries to enable more precise estimation. Finally, early-warning frameworks for journalist safety should extend beyond direct attacks to include the structural conditions identified in this study as amplifiers of risk, such as judicial independence, media ownership concentration, and the electoral trajectory of populist parties. Overall, we cannot predict exactly where or how severely journalist safety will deteriorate, but the direction of risk is clear and worsening, underscoring the need for greater international attention to improving journalist safety worldwide.

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A Appendix A: Chain of Violence – How Populist Regimes Target Journalists

Violence against journalists in populist and authoritarian-populist regimes does not occur in isolation; it operates through a structured and self-reinforcing chain that moves from institutional delegitimisation, through digital hostility and ultimately into physical harm. Understanding this chain requires examining both the mechanisms through which populist governments exert pressure on the press and the concrete patterns of violence that result in systemic silencing of journalists as democratic watchdogs.

According to a UNESCO report (2020), online hostility frequently precedes offline harm. Among

women journalists surveyed, 20% reported physical attacks they believed had been seeded online. The OSCE frames this escalation as a “chain of violence” with three stages. The first is harassment: delegitimisation, online abuse and coordinated attacks that signal to broader society that targeting journalists is acceptable. The second is threats; of rape, murder, or exposure through doxxing, creating a climate of fear and drive self-censorship across the profession. The third stage is physical violence. Critically, when these attacks go unpunished, impunity itself becomes a driver of further violence, creating conditions for increasingly severe harm, including assassination. Violence against journalists in populist regimes is therefore not random. It is sequential, structural, and, in the absence of accountability, self-perpetuating.

B Appendix B: Contextual Factors (Constant Variables)

Military expenditure is used as an indicator of a state’s broader security orientation and institutional priorities. Prior research suggests that higher military spending is often associated with contexts of political instability and weaker institutional safeguards, which can increase risks for journalists (Brambila, 2017). In addition, studies show that when military influence in governance is stronger, state resources may be diverted away from social protection and democratic accountability mechanisms (Shin, 2020). In combination with corruption, higher military expenditure can also contribute to institutional dysfunction, weakening trust in public institutions and reducing protections for civil society actors, including journalists (D’Agostino, Dunne, & Pieroni, 2012).

We also include an indicator of economic development and inequality. Prior research consistently links lower levels of economic development and higher inequality to increased risks of violence against journalists (Brambila, 2017). Schwuchow (2023) further highlights how economic deprivation can be indirectly associated with higher levels of organised crime, which in turn increases exposure to violence targeting the press.

Brambila (2017) identifies judicial independence as a key predictor of journalist killings, as the absence of effective legal recourse increases vulnerability and reduces deterrence against attacks. Where judicial systems lack autonomy, legal protection for journalists is undermined, allowing perpetrators of violence to act with limited accountability.

Finally, government attacks on the judiciary are included as an additional measure of institutional erosion. Aydin-Cakir and Akarçay (2025) show that governments are more likely to target courts under conditions of corruption and political contestation, particularly where independent media can expose wrongdoing. Such attacks reduce judicial checks on power and contribute to an environment in which press freedom is more easily undermined.

C Appendix C: Data Flow Diagram

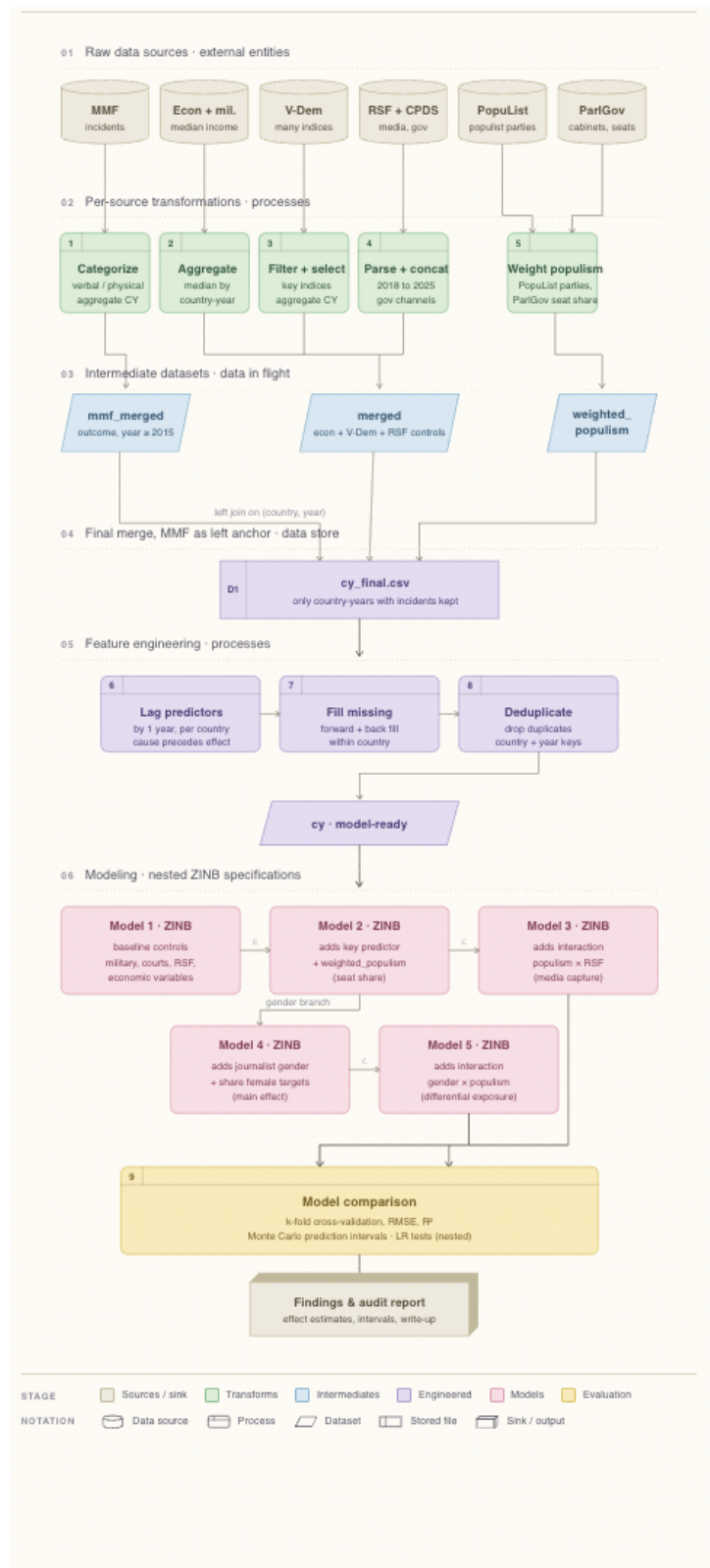


Figure 3: Code Network Analysis of Gendered Threats Against Journalists Across Micro-, Meso-, and Macro-Level Factors

D Appendix D: Annotated Data SourcesModel Development

The three models are built with the datasets as follows:

Model 1:

- Relative poverty: Number of people below 60
- Dataset from Varieties of Democracy (V-dem) (2026): includes 251 indices and 62 other indicators of democracy for 202 states throughout 1789 to 2025.
- Military expenditure (% of GDP) from World Bank (2022): provides the portion of GDP invested in military of countries from 1960 to 2025.

Model 2:

- Weighted populism score: we took variable A from the populist dataset, that assigns a score bla bla bla The PopuList 3.0 dataset (Rooduijn et al., 2023), which offers an overview of populist, far-left, and far-right parties in Europe from 1989 until 2025. PopuList operationalises Mudde’s definition of populism, which is consistent with our theoretical framework. 165 populist parties are identified and only 69 non-populist parties are included.
- The ParlGov dataset, which provides a list of parties, the number of seats they hold, and the total number of seats in cabinet in a given year across 33 European countries and 4 OECD countries ranging from 2015 to 2023.

Model 3: Datasets Variables from model 2

- The World Press Freedom Index, which measures journalists’ ability to report independently and safely, calculated using both incident data and expert assessments for each country (Reporters Without Borders, 2026). To avoid data circularity, we exclude the incident-based component and rely only on the Index’s structural indicators: political, economic, legal, and social conditions.

All the datasets are merged into one final dataframe by country name and year. A one-year lag is applied to predictors relative to the outcome to reduce reverse causality risk and reflect the realistic delay between political conditions and observable journalist violence. Russia and Ukraine are excluded from the analysis, as the violence journalists face in these countries may be reflective of the current conflict rather than a political style.

Missing Values

Missing values are filled by forward-fill within each country group, carrying the value of last year forward to the next if needed. We then assume the majority of remaining missing values, to be missing completely at random (MCAR) and are dropped prior to model training. This assumption holds where gaps in the data reflect administrative inconsistencies or reporting delays rather than any systematic relationship with the outcome variable. Two patterns in the missingness are worth noting. First, incidents of violence against journalists are likely underreported in countries with weaker civil society monitoring infrastructure (An Kwak, 2017), meaning that the countries where press freedom conditions are most severe may paradoxically appear “safer” in the data. This introduces a downward bias in the outcome variable precisely where the model would most benefit from accurate counts. Second, not all countries in the sample have complete coverage across the full 2019–2026 period, which limits the model’s ability to capture temporal trends in

those cases. However, the panel structure retains enough country-year observations to identify meaningful patterns across the sample as a whole.

The final dataframe uses 17 predictors. The following countries are included in the analysis: Austria, Belgium, Bulgaria, Croatia, Cyprus, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovenia, Spain, Sweden, and United Kingdom, spanning from 2019 to 2026 for most countries. This selection was determined by dataset availability after aggregating the datasets.

Evaluation Metrics

The three models are compared to assess the incremental explanatory value of populism and media capture. Model performance is evaluated using a combination of predictive accuracy and goodness-of-fit measures. Across models, comparisons are based on log-likelihood values, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC), with lower values indicating improved model fit. In addition, out-of-sample predictive performance is assessed using root-mean-squared error (RMSE), allowing comparison of how well each model generalises beyond the estimation sample. Together, these metrics allow for a balanced evaluation of model fit and predictive performance. While AIC and BIC assess in-sample explanatory adequacy with a penalty for complexity, RMSE provides an external measure of predictive accuracy. This combined approach ensures that improvements in model fit are not driven solely by overfitting, but reflect genuine gains in explanatory and predictive power.

E Appendix E: Annotated Data Sources

European Centre for Press and Media Freedom (ECPMF). (2026). Mapping Media Freedom [Dataset]. <https://www.mappingmediafreedom.org/> This dataset records individual press freedom violation incidents across European countries, including the type, date, country, and victim characteristics of each alert. For this study, physical incidents include assault, injury, death, abduction, sexual assault, or bombing, in accordance to the ECPMF's categorisation of the dataset incident type, and aggregated to country-year counts. The physical incident count serves as the dependent variable. Data were filtered to 2015 onwards to align with other sources. The dataset provides a granular, incident-level record of journalist violence but relies on reported cases, which may undercount incidents in countries with weaker civil society monitoring infrastructure.

Our World in Data. (2025). Relative poverty: Number of people below 60% of the median [Dataset]. <https://ourworldindata.org/explorers/poverty-wb?Indicator=Number+in+povertyPoverty+line=Relative+poverty%3A+60%25+of+medianHousehold+survey+data+type> Derived from World Bank household survey data, this dataset provides country-year estimates of the number of individuals living below 60% of the national median income, which is a standard EU relative poverty threshold. It is used as a structural economic control to capture the socioeconomic context in which journalist violence occurs, following Brambila's (2017) finding that journalist violence concentrates in areas with high social deprivation. Values are forward-filled within country groups where gaps exist to accommodate the slowly-changing nature of this indicator.

World Bank. (2022). Military expenditure (% of GDP) [Dataset]. World Bank Open

Data. <https://data.worldbank.org/indicator/MS.MIL.XPND.GD.ZS> This dataset provides annual military expenditure as a share of GDP for all World Bank member countries. It is included as a control variable on the basis that high military spending, particularly when combined with institutional corruption, is associated with weakened judicial oversight and increased state-sanctioned violence (D'Agostino, Dunne, & Pieroni, 2012). The data were downloaded in wide format and reshaped to long format before merging. One-year lags are applied to all predictors including this variable to mitigate reverse causality.

Varieties of Democracy (V-dem). (2026). Country-Year: V-Dem Full+Others (Version 16) [Dataset]. Varieties of Democracy (V-Dem). <https://www.v-dem.net/data/the-v-dem-dataset/country-year-v-dem-fullothers-v16/> The V-Dem dataset provides expert-coded indicators of democratic quality at the country-year level. From this source, four variables are extracted: government attacks on the judiciary (v2jupoatk), high court independence (v2juhcind), lower court independence (v2juncind), and media corruption (v2mecorrpt), along with supplementary indicators of civil liberties and social equality. High court independence and government attacks on the judiciary are retained in the final models as controls for the institutional environment structuring journalist safety. The full dataset is large and compressed; only post-2000 observations are retained prior to merging.

Reporters Without Borders. (2026). World Press Freedom Index [Dataset]. <https://rsf.org/en/index> The RSF Press Freedom Index scores countries annually on a 0–100 scale (100 = most free). Due to a methodology change in 2022, two harmonisation approaches are used: for 2018–2021, the pre-built composite score excluding the Safety/exactions component is used; for 2022–2025, the four structural sub-components (Political, Economic, Legal, and Social Context) are averaged, again excluding Safety, to ensure comparability. The resulting score is inverted and rescaled to a media capture index: $(100 - \text{RSF score}) / 100$. This proxy is used in Model 3 as a moderator of the populism–violence relationship, testing whether the effect of populist governance is amplified under conditions of high media capture.

Döring, H., & Manow, P. (2024). ParlGov 2024 release [Dataset]. In Harvard Dataverse. <https://doi.org/10.7910/dvn/2vz5zc> ParlGov provides structured data on parliamentary elections, parties, and cabinet compositions across democratic countries. The cabinet view is used here to extract party-level seat counts and total seats per election, filtered to 2015 onwards. These data are joined with PopuList 3.0 via shared parlgov id identifiers to compute a seat-weighted populism score for each country-year cabinet. The dataset is the methodological bridge between party ideology coding and cabinet-level governance measures.

Rooduijn, M., Pirro, A. L. P., Halikiopoulou, D., Froio, C., Van Kessel, S., De Lange, S. L., Mudde, C., & Taggart, P. (2023). The PopuList (Version V3, Vol. 54, Issue 3, pp. 969–978) [Dataset]. British Journal of Political Science. <https://doi.org/10.1017/s0007123423000431> PopuList 3.0 provides expert classifications of European political parties as populist, far-right, far-left, and/or Eurosceptic. Each party is coded on binary indicators and linked to ParlGov via a shared identifier. In this study, parties coded as populist are matched to their cabinet seat shares from ParlGov to compute a weighted populism score, which is the proportion of governing seats held by populist parties in a given country-year. Parties absent from PopuList are treated as mainstream (coded 0). This operationalisation captures gradations of populist governance rather than treating it as a binary presence/absence variable.

F Appendix F: Description of Results

F.1 Metric Explanation

Coefficient (): The coefficient represents the estimated relationship between a predictor and the outcome variable. In a Zero-Inflated Negative Binomial (ZINB) model, coefficients are estimated on a logarithmic scale, meaning they indicate the direction and strength of the relationship but are not directly interpretable in terms of changes in incident counts

Independent Rate Ratio (IRR): The IRR is obtained by exponentiating the coefficient $IRR = e$. It indicates the multiplicative change in the expected number of incidents associated with a one-unit increase in a predictor. An IRR greater than 1 indicates an increase in the expected number of incidents, while an IRR below 1 indicates a decrease. For example, an IRR of 1.50 implies a 50

p-value: The p-value assesses the statistical significance of a predictor. It represents the probability of observing an effect at least as large as the estimated one if the true effect were actually zero. Smaller p-values provide stronger evidence against the null hypothesis. Conventionally, p-values below 0.05 are considered statistically significant, indicating that the observed relationship is unlikely to be due to chance alone.

F.2 Model 1: Baseline model

Model 1 has a RMSE of 8.12 (± 2.27) and a bias of 0.09 (± 1.27), indicating that the model neither consistently over- nor under-predicts (see Figure1). Although it matches the cross-national pattern of journalist violence fairly well in relative terms, its cross-validated R^2 of 0.01 (± 0.87) is only slightly above zero, indicating that its out-of-sample predictions are only slightly better than a naive model that always predicts the average outcome. Even without any populism measures, the baseline captures the broad cross-national pattern of journalist violence; its prediction error is slightly lower than the outcome's own standard deviation of 10.98.

Relative poverty is the strongest predictor on the full-data refit (coefficient & IRR) ($IRR = 3.27$, $p < .001$) (see figure 3). Approximately 3.25 times the anticipated number of physical incidents is linked to a one-standard-deviation rise in the poverty population. At this point, neither high court independence ($IRR = 0.80$, $p = .21$) nor government attacks on the judiciary ($IRR = 1.23$, $p = .16$) approach significance, and military spending is positive but only marginally significant ($IRR = 1.23$, $p = .096$). This baseline model demonstrates that, prior to the introduction of any populist measures, a country's institutional and socioeconomic structure already explains some portion of the difference in journalist violence, though it does not predict well.

F.3 Model 2: Populism Score

Model 2 returns an RMSE of 9.49 (± 1.69), with a slight inclination to overpredict (bias 0.46 ± 2.02), see Figure 2. The cross-validated R^2 becomes negative (1.89 ± 4.60) and the predicted-observed correlation edges up only marginally to 0.67 (± 0.03), an increase well within fold-to-fold noise (see figure 4). Adding populism, then, adds little predictive power.

Weighted populism is a positive and significant predictor ($IRR = 1.51$, 95% CI [1.15, 1.96], $p = .003$). A one-standard-deviation rise in the seat-weighted populist share in government is associated with approximately 50% more predicted physical incidents (see figure 5). Crucially, populism does not displace the structural predictors but adds to them: relative poverty remains the

dominant single predictor and barely moves from the baseline (IRR = 3.34, $p < .001$, versus 3.27 in Model 1). The introduction of populism also reshapes the institutional coefficients. Government attacks on the judiciary, non-significant in the baseline (IRR = 1.23, $p = .16$), becomes significant here (IRR = 1.39, $p = .024$), a one-standard-deviation rise associating with roughly 39% more incidents, a shift we return to in the model comparison. In the opposite direction, the two security and institutional controls fade once populism enters: military expenditure drops from marginal significance to clearly non-significant (IRR = 1.11, $p = .43$), as does high court independence (IRR = 0.84, $p = .30$).

F.4 Model 3: Media capture as moderating variable

Out-of-sample, Model 3 performs much like Model 2 (see Figure 3. It returns an RMSE of 9.32 (± 1.49), a predicted–observed correlation of 0.67 (± 0.05), and a slight tendency to overpredict (bias 0.41 ± 1.77) (see figure 6). The cross-validated R^2 is again negative on average (1.22 ± 3.18), but the large standard deviation reflects substantial fold-to-fold instability: performance ranges from a respectable R^2 of 0.52 in one fold to 7.57 in another, where the model overpredicts heavily (bias = 3.39).

On the full-data refit, the association between populist governance and physical violence is stronger where media capture is higher as the interactive term variable is a positive and significant predictor (IRR = 1.45, $p = .016$) (see figure 7). Relative poverty remains the dominant predictor (IRR = 3.33, $p < .001$) and government attacks on the judiciary stays significant (IRR = 1.36, $p = .036$), while military expenditure (IRR = 1.07, $p = .64$) and high court independence (IRR = 0.88, $p = .48$) remain non-significant, as in Model 2.

G Appendix G: Stakeholder analysis

Populist government: High power, high hostile interest. The primary independent variable. In countries of high-capture, populist governments are actively and strategically invested in shaping the media landscape to their advantage. This manifests through media capture (ownership pressure, regulatory weaponization, defunding of public broadcasters), delegitimizing rhetoric that frames independent journalists as political enemies, and creating a climate of impunity in which violence goes uninvestigated. Their interest in journalist safety is therefore high but predatory, caring deeply about media, precisely in order to neutralize it. Cabinet gender composition is a sub-variable here.

Military/state: High power, strongly hostile. Represents the coercive apparatus of the state: security services, police, and in some contexts paramilitary actors. In captured or backsliding democracies, these actors can be directly implicated in physical violence against journalists or in failing to investigate and prosecute it (impunity).

Captured outlets: Medium-high power, hostile. Media organizations that have come under political or oligarchic control function as amplifiers of anti-press narratives rather than defenders of journalistic safety. Captured outlets are both a consequence of media capture and a reinforcing mechanism, normalizing hostility toward independent colleagues and reducing the perceived social cost of attacks. Their position in this quadrant is itself partly an outcome of populist governments' active media strategy.

Independent outlets: Medium power, protective. Structurally opposed to captured outlets. They have genuine interest in journalist safety but limited institutional power, especially in

high-capture environments. Their position in the upper-right reflects that they are allies with meaningful organizational capacity (platforms, visibility, solidarity networks) but are increasingly squeezed by political and economic pressure.

EU Institutions: High power, protective ally. Represent the supranational regulatory and normative framework like the European Commission, Parliament, and mechanisms like the Media Freedom Act and Rule of Law reports. High interest in journalist safety as part of democratic backsliding concerns, and significant structural power via conditionality, infringement proceedings, and funding leverage. However, their practical effectiveness is contested, particularly in member states with entrenched populist governance

RSF/UNESCO: Medium-high power, strongly protective. Reporters Without Borders and UNESCO operate as the key international monitoring and norm-setting bodies. They have high interest and meaningful soft power (naming-and-shaming, international visibility, standard-setting) but no enforcement capacity.

Press-freedom NGOs: Lower-medium power, strongly protective. The broader ecosystem of civil society organizations below RSF/UNESCO in institutional weight, being national and regional press freedom organizations, journalist associations, and legal defense funds. They are the closest actors to journalists on the ground, providing documentation, legal aid, and psychosocial support, but operate with limited resources and are often targeting themselves in high-capture environments.

Journalists: Low institutional power, strongly protective (of themselves and peers). The subjects of your study and the most directly affected stakeholders. Positioned low on the power axis because individually they have little structural leverage, especially in hostile environments where self-censorship (a key outcome variable in your model) is the rational response. Their high interest represents how safety is existential for them. Their vulnerability is also shaped by gender, beat, and employment precarity.

H Appendix H: Participant Overview

Table 7: Overview of interview participants

Gender	Country of work	Position
Male	Belgium	Data Journalist
Female	Bulgaria	Fact Checker
Female	Czech Republic	Political History Journalist
Female	Portugal	News Intern
Female	Netherlands	Student Journalist
Female	England	Freelance Journalist

I Appendix I: Interview guide

Section A - Background & Context – Goal: Situate the participant [Start of with a brief explanation of our topic]

- Can you briefly describe who you are (+ where you are from) and what you do now?

- Can you describe your role and experience?
- How does your outlet's ownership, funding, or national context shape what can and cannot be said?

Section B – Experiences with hostility / intimidation, self-censorship

- How would you describe the current climate for journalists in [country/Europe]?
- Have you observed or experienced forms of hostility or violence towards journalists?
 - Who are the typical actors? / Who were the actors?
 - Could you tell us a bit more about the hostility you experienced?
 - What was the situation? (p.e. offline/online)
 - If yes- what do you think is the reason for the hostility?
- Have you changed your reporting behaviour: what you cover, how visible you are online, whether you use a byline — because of safety concerns? What drove that decision?

Section C – Role of political leadership (populism)

- Are you familiar with populism?
- Some populist leaders explicitly frame mainstream media as an "enemy." Have you experienced that hostility?
- How much do you believe it has depended on your gender identity?

Section D – Gender and political culture

- Do you think gender dynamics in governing bodies affect public perception of journalists, and why?
 - How is your work environment affected by these dynamics?
 - How would you expect male-dominated governance to influence the treatment of journalists
 - Can you think of specific examples from your own experience?
 - Can you describe a moment where you felt your gender influenced how a source, official, or colleague treated you in a professional context?
 - Have you ever felt that attacks on your credibility were framed around your gender rather than your journalism — e.g., being called emotional, biased, or ideologically driven in gendered terms?

E – Postcolonialism & Miscellaneous

- How does your own background — national, ethnic, linguistic — shape the stories you're drawn to, or the access you're granted or denied?
- Do you feel your editorial judgments are trusted equally to those of colleagues with different backgrounds? Can you give an example?
- Are there stories you've pitched that were rejected in ways that felt tied to your background or perspective rather than news value?

Section F – Reflection & Interpretation

- Overall, do you believe that the previous topics (gender, political culture, and post colonialism ties) have an indirect effect of hostility towards journalists by the government?
 - How do you think this would differ if a journalist lived in an area with a populist governance?
- Do you think our current explanations miss anything important? Are there any structural powers that you think exist in the news industry not mentioned?
 - If yes- why do you think that is? How were these structural imbalances of power put into place? When?

J Appendix I: Complete Bias Overview

The interviews surfaced seven interconnected factors listed below that shape whether journalists can operate freely and safely.

Political polarisation emerged as a structural constraint on editorial courage. In societies where topics are perceived as increasingly divisive, outlets become more cautious in how they frame and present sensitive issues, which is a form of self-censorship in of itself that is both a product of existing culture and a force that reinforces it. Left unchecked, this dynamic gradually erodes the conditions under which journalists can genuinely function as democratic watchdogs.

Media trust operates as a cultural precondition for journalistic access. Where public trust in the media has deteriorated, people become less willing to speak to journalists, narrowing the plurality of narratives that healthy journalism depends on. This distrust also deepens the Manichaeian divide between journalists and the public, casting the press as part of a hostile elite rather than a public service, and in doing so, adds fuel to cultures of hostility against the profession.

Ownership and funding structures determine the practical boundaries of editorial independence. Outlets that rely on a single dominant sponsor face implicit pressure to avoid content that might jeopardise that relationship, even in the absence of explicit censorship. Diversified funding across readers, grants, and institutional sources is what protects the freedom to write without calculating the cost to a key funder.

Gender hierarchies impose an additional layer of professional delegitimisation in societies shaped by macho cultural norms. Women journalists in these environments are rarely perceived as journalists first, rather, they are perceived as women journalists, which is a distinction that carries real consequences for how seriously their work is taken, how they are treated by sources and institutions, and how vulnerable they are to targeted hostility. Moreover, when youth compounds gender, the burden compounds further, putting young women journalists up to face a dual credibility deficit that can fundamentally alter the environments in which they work and the stories they are able to pursue.

Attacks on journalistic expertise follow a similar logic for younger journalists regardless of gender. When a journalist's age is used to dismiss their authority, it reshapes the professional dynamic before any reporting has taken place. The journalist is not engaged with as a journalist, but as a young journalist, a framing that positions them as less credible, less threatening, and

therefore easier to dismiss as a real viability of keeping democracy in check.

Online sexual harassment constitutes a distinctly gendered form of attack that falls disproportionately on young women journalists. Beyond its immediate psychological impact, it functions as a mechanism of exclusion making certain spaces, topics, and sources feel inaccessible or unsafe, and contributing to the broader chilling effect that can potentially shape what journalists choose to cover.

The **anonymity of attackers** affects how seriously journalists are able to take threats and hostile behaviour. When an attack cannot be traced to an identifiable source, journalists frequently rationalise it as noise, such as a bot, a troll, or an irrelevant provocation, rather than a credible threat. This tendency, while understandable, risks underestimating the cumulative psychological weight of sustained anonymous hostility and the organised nature of some coordinated attack campaigns.

Taken together, these factors map directly onto the gap between the ideal journalistic environment described and the lived reality of most journalists across our interviews. They reveal that what prevents journalism from fulfilling its democratic function is rarely a single dramatic intervention, but rather the slow accumulation of structural pressures and cultural changes that quietly reshape what journalists feel able to do, which can potentially fuel an environment of hate and violence.

K Statements

K.1 Failure Analysis statement

K.1.1 LLM-Labeling

We attempted to implement LLM-based labelling of perpetrators of violence against journalists based on online articles sentiments, which was highly costly both computationally and financially and ultimately did not produce usable results: lots of missing data and was not feasible to fact-check a reasonable amount of outputs considering how many rows it produced. We tried a pro-regime classifier as well as a populist score classifier which can be found in the code.

K.1.2 Chain of Violence

We could not operationalise the escalation of violence that we originally hypothesised in our initial outline for the project as it contrasted with the rest of our methodology and modelling constraints.

K.2 Annotated list of relevant laws and regulations for the jurisdiction

- European Media Freedom Act
 - The European Media Freedom Act was put into force on 7 May 2024 and includes a set of rules to protect media pluralism and independence in the EU. Such rules include enhancing transparency of ownership, protecting editorial independence, ensuring the independent function of the media, etc.
- EU Artificial Intelligence Act
 - The use of AI systems shall comply with the requirements of the EU Artificial Intelligence Act and all applicable data protection, privacy, and fundamental rights legislation. AI

systems must be developed and deployed in a manner that is transparent, safe, and subject to human oversight. Regular monitoring and evaluation shall be conducted to identify and mitigate risks related to bias, discrimination, security, and unintended impacts to groups.

- General Data Protection Regulation

- The General Data Protection Regulation (GDPR) establishes rules for the fair and transparent processing of personal data within the European Union. AI systems that process personal data must comply with GDPR requirements, including data minimisation, accuracy, security, and accountability. Individuals retain rights over their personal data, including access and the possibility to delete their information, and protection against certain forms of automated decision-making. Organisations using AI must ensure that personal data is handled responsibly and that appropriate safeguards are in place to protect privacy rights.

K.3 Statement of Project Work Division

<https://docs.google.com/document/d/1hkE8EZ1ofUYGIhcxztgAiyIUwVdp3pcM/edit?usp=sharing&ouid=116609130675846779758&rtpof=true&sd=true>

K.4 Statement of Project Outline Documentation

https://docs.google.com/document/d/1zsRxft1ZxyZE5_wgUaylS1dgJvFdRQnf7xvpm2aRizA/edit?usp=sharing